

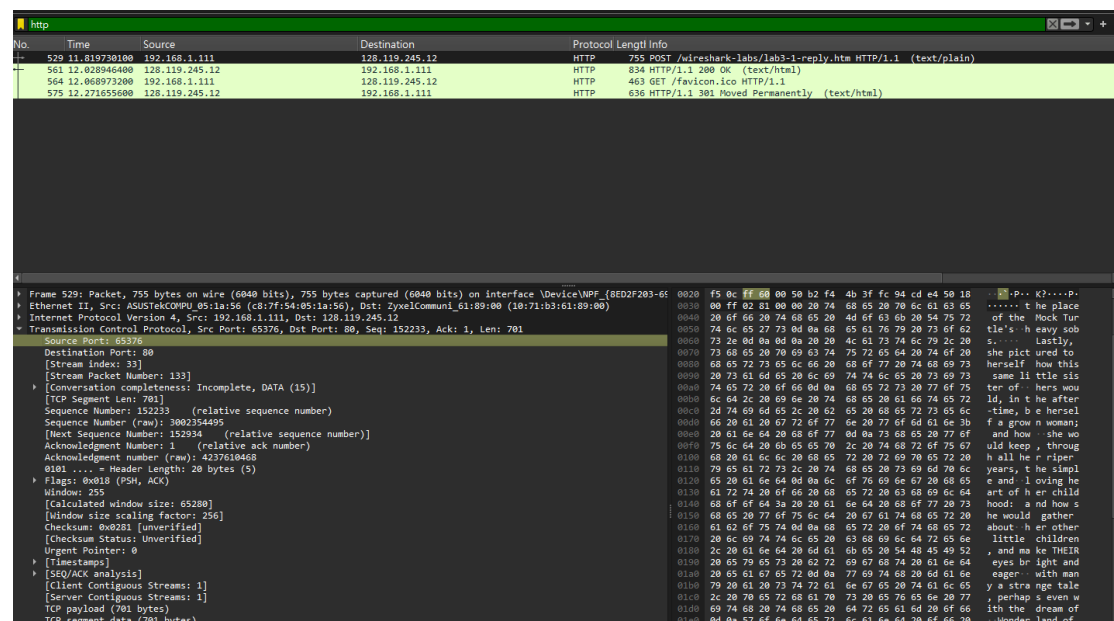
Part 1. Capturing a bulk TCP transfer from your computer to a remote server (無題目，僅上傳文字檔)

Part 2. A first look at the captured trace (第 1~2 題)

1. What is the IP address and TCP port number used by the client computer (source) that is transferring the alice.txt file to gaia.cs.umass.edu? To answer this question, it's probably easiest to select an HTTP message and explore the details of the TCP packet used to carry this HTTP message, using the “details of the selected packet header window” (refer to Figure 2 in the “Getting Started with Wireshark” Lab if you're uncertain about the Wireshark windows).

IP 位址：192.168.1.111

TCP Port：65376



2. What is the IP address of gaia.cs.umass.edu? On what port number is it sending and receiving TCP segments for this connection?

IP 位址：128.119.245.12

TCP Port：80

Part 3. TCP Basics (第 3~10 題)

3. What is the *sequence number* of the TCP SYN segment that is used to initiate the TCP connection between the client computer and gaia.cs.umass.edu? (Note: this is the “raw” sequence number carried in the TCP segment itself; it is *NOT* the packet # in the “No.” column in the Wireshark window.) What is it in this TCP segment that identifies the segment as a SYN segment? Will the TCP receiver in this session be able to use Selective Acknowledgments?

Sequence Number (raw): 64353152

Flags: 0x002 (SYN)

TCP Option - SACK permitted Kind: SACK Permitted (4) Length: 2 => YES

Wireshark packet capture showing a TCP SYN segment from 192.168.1.111 to 128.119.245.12. The packet details show the raw sequence number 64353152 and the SYN flag set. The packet bytes show the raw sequence number 64353152 in the TCP header.

Packet 128: Frame 128: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF... (8ED2F203-6958-4968-B59C-SA61A69E3) Ethernet II, Src: ASUSTekCOMP05:1a:56 (c8:7f:54:05:1a:56), Dst: ZyxelCommuni_61:89:00 (10:71:b3:61:89:00) Internet Protocol Version 4, Src: 192.168.1.111, Dst: 128.119.245.12 Transmission Control Protocol, Src Port: 65169, Dst Port: 443, Seq: 0, Len: 0

Source Port: 65169
Destination Port: 443
[Stream index: 13]
[Stream Packet Number: 1]
[Conversation completeness: Complete, WITH_DATA (63)]
[TCP Segment Len: 0]
Sequence Number: 0 (relative sequence number)
Sequence Number (raw): 64353152
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 0
Acknowledgment number (raw): 0
1800 = Header Length: 32 bytes (8)
Flags: 0x002 (SYN)
...0... = Reserved: Not set
...0... = Accurate ECN: Not set
...0... = Congestion Window Reduced: Not set
...0... = ECH: Echo: Not set
...0... = Urgent: Not set
...0... = Acknowledgment: Not set
...0... = Push: Not set
...0... = Reset: Not set
...1... = SYN: set
...0... = Fin: Not set
[TCP Flags:S.]
Window: 65535
[Calculated window size: 65535]
Checksum: 0x3fd2 [unverified]
[Checksum Status: Unverified]
Urgent pointer: 0
Options: (12 bytes), Maximum segment size, No-Operation (NOP), Window scale, No-Operation (NOP), No-Operation (NOP), SACK permitted
TCP Option - Maximum segment size: 1460 bytes
TCP Option - No-Operation (NOP)
TCP Option - Window scale: 8 (multiply by 256)
TCP Option - No-Operation (NOP)
TCP Option - No-Operation (NOP)
TCP Option - SACK permitted
Kind: SACK Permitted (4)
Length: 2
[Timestamps]
[Client Contiguous Streams: 1]
[Server Contiguous Streams: 1]

4. What is the *sequence number* of the SYNACK segment sent by gaia.cs.umass.edu to the client computer in reply to the SYN? What is it in the segment that identifies the segment as a SYNACK segment? What is the value of the Acknowledgement field in the SYNACK segment? How did gaia.cs.umass.edu determine that value?

Sequence Number (raw): 55775529

如何辨識此封包為 **SYNACK** 封包？Flags Syn = Set 或是 Flags = 0x012

```
133 1.827497200 128.119.245.12 192.168.1.111 TCP 66 443 → 65169 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1452 SACK_PERM=128
134 1.827675400 192.168.1.111 128.119.245.12 TCP 54 65169 → 443 [ACK] Seq=1 Ack=1 Win=65200 Len=0
135 1.828973800 192.168.1.111 128.119.245.12 TCP 1586 65169 → 443 [ACK] Seq=1 Ack=1 Win=65200 Len=1452 [TCP PDU reassembled in 136]
136 1.828973800 192.168.1.111 128.119.245.12 TLSv1.3 361 Client Hello (SNI=gaia.cs.umass.edu)
137 2.046988500 128.119.245.12 192.168.1.111 TCP 60 443 → 65169 [ACK] Seq=1 Ack=1760 Win=62592 Len=0
138 2.046988500 128.119.245.12 192.168.1.111 TLSv1.3 1586 Server Hello, Change Cipher Spec, Application Data
139 2.0470818200 128.119.245.12 192.168.1.111 TCP 1586 443 → 65169 [PSH, ACK] Seq=1453 Ack=1760 Win=62592 Len=1452 [TCP PDU reassembled in 140]
140 2.0470823200 128.119.245.12 192.168.1.111 TLSv1.3 700 Application Data, Application Data, Application Data

4 Frame 133: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{BED2F203-6958-4968-B59C-SA61A69E3...}
  Ethernet II, Src: ZyxelCommuni_01:89:00 (18:71:b5:01:89:00), Dst: ASUSTekCOMPU_05:1a:56 (c8:7f:94:05:1a:56)
  Internet Protocol Version 4, Src: 128.119.245.12, Dst: 192.168.1.111
  Transmission Control Protocol, Src Port: 443, Dst Port: 65169, Seq: 0, Ack: 1, Len: 0
    Source Port: 443
    Destination Port: 65169
    [Stream Index: 13]
    [Stream Packet Number: 2]
    [Conversation completeness: Complete, WITH_DATA (63)]
    [TCP Segment Len: 0]
    Sequence Number: 0 (relative sequence number)
    Sequence Number (raw): 55775529
    [Next Sequence Number: 1 (relative sequence number)]
    Acknowledgment Number: 1 (relative ack number)
    Acknowledgment number (raw): 64353153
    1000 .... = Header Length: 32 bytes (0)
    Flags: 0x012 (SYN, ACK)
      0000 .... = Reserved: Not set
      ...0 .... = Accurate ECN: Not set
      ....0. .... = Congestion Window Reduced: Not set
      ....0. .... = ECH-Echo: Not set
      ....0. .... = Urgent: Not set
      ....1. .... = Acknowledgment: Set
      ....0.. = Push: Not set
      ....0.. = Reset: Not set
    ... .. = Syn Set
    ... .. = Fin: Not set
    [TCP Flags: .....A..S.]
    Window: 64240
    [calculated window size: 64240]
    Checksum: 0x385d [unverified]
    [checksum status: Unverified]
    Urgent Pointer: 0
    Options: (12 bytes), Maximum segment size, No-Operation (NOP), No-Operation (NOP), SACK permitted, No-Operation (NOP), Window scale
      TCP Option - Maximum segment size: 1452 bytes
      TCP Option - No-Operation (NOP)
      TCP Option - No-Operation (NOP)
      TCP Option - SACK permitted
        Kind: SACK Permitted (4)
        Length: 2
      TCP Option - No-Operation (NOP)
      TCP Option - Window scale: 7 (multiply by 128)
      [Timestamps]
      [SEQ/ACK analysis]
      [Client Contiguous Streams: 1]
      [Server Contiguous Streams: 1]
```

Acknowledgment number (raw): 64353153

伺服器 (gaia.cs.umass.edu) 如何決定此數值？：伺服器是將接收到的 SYN 封包原始序號直接 加 1 得到的。這個「+1」的動作代表伺服器已成功收到連線請求，並期待 Client 下一次發送序號從接續號碼開始的資料。

否，alice.txt 148 KB

No.	Time	Source	Destination	Protocol	Length	Info
529	11.819730100	192.168.1.1111	128.119.245.12	HTTP	755	POST /w/reshark-lab3/lab3-1-reply.htm HTTP/1.1 (text/plain)
<pre>Frame 529: Capture, 755 bytes on wire (6040 bits), 755 bytes captured (6040 bits) on interface \Device\NPF{...} Ethernet II, Src: ASUSTekCompu05:1a:56 (c8:7f:54:05:1a:56), Dst: ZyxellCommuni 61:89:00 (10:71:b3:61:89:00) Internet Protocol Version 5, Src: 192.168.1.1111, Dst: 128.119.245.12 Transmission Control Protocol, Src Port: 65376, Dst Port: 80, Seq: 152233, Ack: 1, Len: 701 Source Port: 65376 Destination Port: 80 [Stream index: 33] [Stream Packet Number: 133] [Conversation completeness: Incomplete, DATA (15)] [TCP Segment Len: 701] Sequence Number: 152233 (relative sequence number) Sequence Number (raw): 3002354495 [Next Sequence Number: 152934 (relative sequence number)] Acknowledgment Number: 1 (relative ack number) Acknowledgment number (raw): 4237610468 0101 = Header Length: 20 bytes (5) * Flags: 0x01B (PSH, ACK) 0... = Reserved: Not set 0... = Accurate ECN: Not set 0... = Congestion Window Reduced: Not set 0... = ECH-Echo: Not set 0... = Urgent: Not set 1... = Acknowledgment: Set 1... = Push: Set 0... = Reset: Not set 0... = SYN: Not set 0... = FIN: Not set [TCP Flags:]AP... Window: 255 [Calculated window size: 65308] [Window size scaling factor: 256] Checksum: 0x0281 [unverified] [Checksum Status: Unverified] Urgent Pointer: 0 [Timestamp] [SEQ/ACK analysis] [Client Contiguous Streams: 1] [Server Contiguous Streams: 1] TCP payload (701 bytes) TCP segment data (701 bytes) (107) Reassembled TCP Segments (152933 bytes): #390(612), #391(1452), #392(1452), #393(1452), #394(1452), #395(1452), #396(1452), #397(1452) Hypertext Transfer Protocol POST /w/reshark-lab3/lab3-1-reply.htm HTTP/1.1/\r\n Host: gata.cs.umass.edu/\r\n Connection: keep-alive/\r\n Content-Length: 152321/\r\n Cache-Control: max-age=0/\r\n Upgrade-Insecure-Requests: 1/\r\n User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36/\r\n Origin: null/\r\n Content-Type: multipart/form-data; boundary=-----WebKitFormBoundaryn1xhDF3-B8u1e/\r\n Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.0/\r\n Accept-Encoding: gzip, deflate/\r\n Accept-Language: zh-TW,zh;q=0.9/\r\n \r\n</pre>						
00000	10.71.b3.61.89.00	c8.7f.54.05.1a.56	08.00.45.00.00	45.00	00	q a c t i v e
00110	02.55.4d.4e.40.80	89.6e.08.c8.a8	01.6f.80.77	00.00	00	ing...s...ow
00200	fs.0c.f7.60.00.50	b2.f4.4b.3f.c4	cd.4e.50.18	00.00	00	g...K3...p...
00300	00.0f.02.81.00.00	20.74.68.65	20.70.6c.61.63.65	00.00	00	the place
00400	20.6f.66.20.74.68.65	20.4d.6f.63.68.20.54	79.72.6f.72.6f.72.6f.63.65	00.00	00	...the Mock Tur
00500	74.6c.65.27.73.0d.0a.08	05.61.70.79.20.73.6f.63.65	00.00	00	00	tie's...heavy sob
00600	73.2e.0d.0a.0d.0a.20.4c	61.73.74.6c.79.2c.20	00.00	00	00	s...Lastly,
00700	73.68.65.20.70.69.63.74	75.72.65.64.20.74.6f.20	00.00	00	00	she pictured to
00800	68.65.72.73.65.6c.66.20.68	6f.77.20.74.68.69.73	00.00	00	00	herself...how this
00900	20.73.61.6d.65.20.6c.69	74.74.6c.65.20.73.69.73	00.00	00	00	same little s
00a00	74.65.72.6f.6f.6d.0a.08	68.65.72.73.20.77.6f.75	00.00	00	00	ter of...hers wou
00b00	6c.64.2c.20.69.2e.20.74	68.65.20.61.66.74.75.65.62	00.00	00	00	ld, in the after
00c00	20.74.69.6d.65.2c.20.62	65.20.68.65.72.73.6f.6c	00.00	00	00	-time, be hersel
00d00	68.20.61.20.67.72.6f.77	6e.20.77.6f.6d.61.6c.6e.30	00.00	00	00	f a grown woman,
00e00	20.61.6e.64.20.68.6f.77	0d.0a.73.68.65.20.77.6f.6f	00.00	00	00	and how...she wo
00f00	75.6c.64.20.6b.65.65.70	2c.20.74.68.72.6f.75.6f.75	00.00	00	00	uld keep, throug
01000	68.20.61.6e.64.20.68.65	72.72.62.79.70.65.72.20	00.00	00	00	h all her riper
01100	70.65.61.72.73.2c.20.74	68.65.20.73.69.6d.61.6c.6e.30	00.00	00	00	years, the simpl
01200	65.20.61.6e.64.0d.6a.0c	6f.76.69.6f.20.62.68.65.65	00.00	00	00	e and...loving he
01300	61.72.72.20.6f.66.20.68	65.72.20.63.68.69.6c.64	00.00	00	00	art of her child
01400	68.6f.6f.64.3a.20.20.61	6e.64.20.68.6f.77.20.73	00.00	00	00	hood;...and how s
01500	68.65.20.77.6f.75.6c.64	20.67.61.74.68.65.72.20	00.00	00	00	he would gather
01600	61.62.6f.75.74.0d.0a.08	65.72.20.74.68.65.72.00	00.00	00	00	about...her othe
01700	20.6c.69.74.74.6c.65.20	63.68.69.6c.64.72.65.6e	00.00	00	00	little children
01800	2c.20.61.6e.64.20.6d.61	68.65.20.54.48.45.49.61.62	00.00	00	00	, and make THEIR
01900	65.20.79.65.73.20.02.72	69.67.68.74.20.61.6c.64	00.00	00	00	eyes;br...ight and
01a00	20.65.61.6f.65.72.0d.0a	77.69.74.68.69.6d.61.6c.6e.30	00.00	00	00	eager...with man
01b00	79.20.61.20.73.74.72.61	6e.67.65.20.74.61.6c.65	00.00	00	00	y a strange tale
01c00	2c.20.70.65.72.68.61.70	73.20.65.76.65.6e.20.77	00.00	00	00	, perhaps s even w
01d00	70.64.82.20.74.68.65.20	64.72.65.61.6d.20.6f.66	00.00	00	00	ith the dream of
01e00	8d.0a.57.6f.64.65.72.6c	61.61.6c.64.20.6f.68.20	00.00	00	00	the...surreal
01f00	6c.6f.6e.67.20.61.61.6f.67	3a.20.61.6e.61.64.20.68	00.00	00	00	long ago;...and h
02000	6f.77.20.73.68.65.20.77	67.65.6c.64.20.69.65.65	00.00	00	00	ow she w ould fee
02100	6c.20.77.69.74.68.65.20.61	6c.64.20.74.68.69.65.72	00.00	00	00	l with a ll their
02200	8d.0a.73.6d.6f.70.6c.65	20.73.6f.72.72.6f.77.20	00.00	00	00	l...surreal
02300	2c.20.61.6e.64.20.66.69	6e.64.20.61.20.70.6c.65	00.00	00	00	, and find a ple
02400	61.73.75.72.65.20.69.6e	20.61.6c.62.20.74.68.65	00.00	00	00	asure in all the
02500	69.72.20.73.69.6d.70.6c	6c.65.20.6f.6f.73.73.2c.0d	00.00	00	00	in simple joys,
02600	0a.72.65.6d.6d.62.65.72	69.6f.6e.6f.20.68.65.65.2c	00.00	00	00	, newborn...gag her
02700	20.6f.77.6e.20.63.68.69	6e.64.20.6f.69.65.65.2c	00.00	00	00	child-life, the
02800	20.61.6e.64.20.74.68.65	20.68.61.70.79.70.73	00.00	00	00	and the happy s
02900	75.6d.65.72.20.64.61.79	73.20.65.74.0d.0a.08.20	00.00	00	00	ummer da ys...
03000	20.20.20.20.20.20.20.20	20.20.20.20.20.20.20.20	00.00	00	00	
03100	20.20.20.20.20.20.20.20	20.20.20.20.20.20.20.20	00.00	00	00	
03200	45.4e.44.0d.0a.0d.0a.2d	2d.2d.2d.2d.2d.2d.2d.2d	00.00	00	00	END -----THE
03300	40.69.74.46.6f.72.6d.42	6f.75.6e.64.71.62.79.4c	00.00	00	00	Kittiforme oundar
03400	2e.69.78.68.6c.44.46.33	72.42.73.6b.65.45.62.2d	00.00	00	00	nixhDf3 r8skubk-
03500	2d.0d.0a		00.00	00	00	

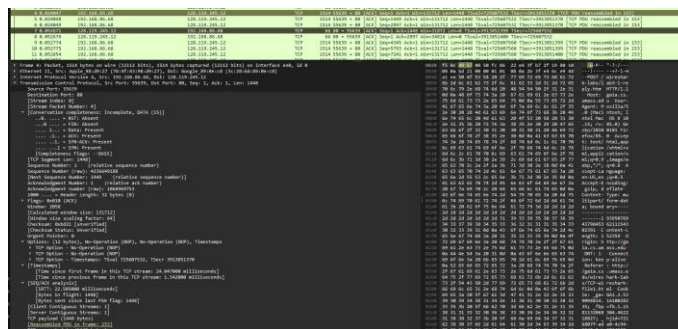
Packet (755 bytes) Reassembled TCP (152933 bytes)

(第六題開始找不到答案，前面用自己抓的，這邊開始用 **tcp-wireshark-trace1-1.pcapng** 寫答案)

6. Consider the TCP segment containing the HTTP “POST” as the first segment in the data transfer part of the TCP connection.

- At what time was the first segment (the one containing the HTTP POST) in the data-transfer part of the TCP connection sent?

0.024047 秒 (看編號 #4 封包)。



- At what time was the ACK for this first data-containing segment received?

0.052671 (看編號 #7 封包，其 Ack=1449 正好是針對 #4 的確認)。

- What is the RTT for this first data-containing segment?

計算：Frame 7 時間 0.052671 - Frame 4 時間 0.024047 = 0.028624 秒

- What is the RTT value the second data-carrying TCP segment and its ACK?

第二筆送出時間 (#5)：0.024048。其對應 ACK 時間 (#8)：0.052676。

計算結果：0.052676 - 0.024048 = 0.028628 秒。

- What is the EstimatedRTT value after the ACK for the second data-carrying segment is received? Assume that in making this calculation after the received of the ACK for the second segment, that the initial value of EstimatedRTT is equal to the measured RTT for the first segment, and then is computed using the EstimatedRTT equation on page 242, and a value of $\alpha = 0.125$.

EstimatedRTT = $0.875 \times 0.028624 + 0.125 \times 0.028628 = 0.028624$ 秒

7. What is the length (payload) of each of the first four data-carrying TCP segments?

1448, 1448, 1448, 1448 bytes (都一樣)

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.86.68	128.119.245.12	TCP	78	55639 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=725607509 TSecr=0 SACK_PERM
2	0.022414	128.119.245.12	192.168.86.68	TCP	74	80 → 55639 [SYN, ACK] Seq=0 Ack=1 Win=28960 Len=0 MSS=1460 SACK_PERM TSval=3913851370 TSecr=725607509 WS=128
3	0.022585	192.168.86.68	128.119.245.12	TCP	66	55639 → 80 [ACK] Seq=1 Ack=1 Win=131712 Len=0 TSval=725607531 TSecr=3913851370
4	0.024847	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=1 Ack=1 Win=131712 Len=1448 TSval=725607532 TSecr=3913851370 [TCP PDU reassembled in 153]
5	0.024848	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=1449 Ack=1 Win=131712 Len=1448 TSval=725607532 TSecr=3913851370 [TCP PDU reassembled in 153]
6	0.024849	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=2897 Ack=1 Win=131712 Len=1448 TSval=725607532 TSecr=3913851370 [TCP PDU reassembled in 153]
7	0.052671	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=1449 Win=31872 Len=0 TSval=3913851399 TSecr=725607532
8	0.052676	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=2897 Win=34816 Len=0 TSval=3913851400 TSecr=725607532
9	0.052774	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=4345 Ack=1 Win=131712 Len=1448 TSval=725607560 TSecr=3913851399 [TCP PDU reassembled in 153]

8. What is the minimum amount of available buffer space advertised to the client by gaia.cs.umass.edu among these first four data-carrying TCP segments? Does the lack of receiver buffer space ever throttle the sender for these first four data-carrying segments?

看 server 回傳的 ACK 中的 Win 欄位（前四個資料段對應的 ACK）：

7 0.052671 128.119.245.12 192.168.86.68 TCP 66 80 → 55639 [ACK]

Seq=1 Ack=1449 Win=31872 Len=0 TSval=3913851399 TSecr=725607532

8 0.052676 128.119.245.12 192.168.86.68 TCP 66 80 → 55639 [ACK]

Seq=1 Ack=2897 Win=34816 Len=0 TSval=3913851400 TSecr=725607532

13 0.053626 128.119.245.12 192.168.86.68 TCP 66 80 → 55639 [ACK]

Seq=1 Ack=4345 Win=37760 Len=0 TSval=3913851400 TSecr=725607532

16 0.080768 128.119.245.12 192.168.86.68 TCP 66 80 → 55639 [ACK]

Seq=1 Ack=5793 Win=40576 Len=0 TSval=3913851421 TSecr=725607560

最小 **buffer = 31872 bytes**

有沒有被 **throttle**？沒有

- 理由：Frame 4-6 在 0.024 秒立即發送
- Frame 7-8 的 ACK 一到（0.052 秒），Frame 9-12 馬上繼續發送
- 說明 buffer 充足，sender 沒被限制

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.86.68	128.119.245.12	TCP	78	55639 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=725607509 TSecr=0 SACK_PERM
2	0.022414	128.119.245.12	192.168.86.68	TCP	74	80 → 55639 [SYN, ACK] Seq=0 Ack=1 Win=28960 Len=0 MSS=1460 SACK_PERM TSval=3913851370 TSecr=725607509 WS=128
3	0.022585	192.168.86.68	128.119.245.12	TCP	66	55639 → 80 [ACK] Seq=1 Ack=1 Win=131712 Len=0 TSval=725607531 TSecr=3913851370
4	0.024847	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=1 Ack=1 Win=131712 Len=1448 TSval=725607532 TSecr=3913851370 [TCP PDU reassembled in 153]
5	0.024848	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=1449 Ack=1 Win=131712 Len=1448 TSval=725607532 TSecr=3913851370 [TCP PDU reassembled in 153]
6	0.024849	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=2897 Ack=1 Win=131712 Len=1448 TSval=725607532 TSecr=3913851370 [TCP PDU reassembled in 153]
7	0.052671	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=1449 Win=31872 Len=0 TSval=3913851399 TSecr=725607532
8	0.052676	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=2897 Win=34816 Len=0 TSval=3913851400 TSecr=725607532
9	0.052774	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=4345 Ack=1 Win=131712 Len=1448 TSval=725607560 TSecr=3913851399 [TCP PDU reassembled in 153]
10	0.052775	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=5793 Ack=1 Win=131712 Len=1448 TSval=725607560 TSecr=3913851399 [TCP PDU reassembled in 153]
11	0.052854	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=7241 Ack=1 Win=131712 Len=1448 TSval=725607560 TSecr=3913851400 [TCP PDU reassembled in 153]
12	0.052855	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=6069 Ack=1 Win=131712 Len=1448 TSval=725607560 TSecr=3913851400 [TCP PDU reassembled in 153]
13	0.053626	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=4345 Win=37760 Len=0 TSval=3913851400 TSecr=725607532
14	0.053710	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=10137 Ack=1 Win=131712 Len=1448 TSval=725607560 TSecr=3913851400 [TCP PDU reassembled in 153]
15	0.053731	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=11585 Ack=1 Win=131712 Len=1448 TSval=725607560 TSecr=3913851400 [TCP PDU reassembled in 153]
16	0.080768	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=5793 Win=40576 Len=0 TSval=3913851421 TSecr=725607560

9. Are there any retransmitted segments in the trace file? What did you check for (in the trace) in order to answer this question?

沒有重傳

檢查方法：

1. 看有沒有 [TCP Retransmission] 標記
2. 看有沒有 相同 Seq number 出現多次
3. 從 trace file 看，所有 Seq number 都是遞增的，沒有重複

10. What is the throughput (bytes transferred per unit time) for the TCP connection? Explain how you calculated this value.

看 trace file 中的時間和傳輸資料量：

傳輸時間 = 0.191496 - 0.024047 = 0.167449 秒

傳輸資料 = 153426 bytes

Throughput = 153426 bytes / 0.167449 秒 = 915,825 bytes/秒

≈ 915.8 KB/s 或 7.3 Mbps

[第一個數據段開始發送]

4 0.024047 192.168.86.68 128.119.245.12 TCP 1514 55639 → 80 [ACK]
Seq=1 Ack=1 Win=131712 Len=1448 TSval=725607532 TSecr=3913851370 [TCP PDU
reassembled in 153]

4	0.024047	192.168.86.68	128.119.245.12	TCP	1514	55639 → 80 [ACK] Seq=1 Ack=1 Win=131712 Len=1448 TSval=725607532 TSecr=3913851370 [TCP PDU reassembled in 153]
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[server 最後一個 ACK]

178 0.191496 128.119.245.12 192.168.86.68 TCP 66 80 → 55639 [ACK] Seq=1
Ack=153426 Win=194816 Len=0 TSval=3913851541 TSecr=725607650

178	0.191496	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK] Seq=1 Ack=153426 Win=194816 Len=0 TSval=3913851541 TSecr=725607650
179	0.192625	128.119.245.12	192.168.86.68	HTTP	843	HTTP/1.1 200 OK (text/html)
180	0.192732	192.168.86.68	128.119.245.12	TCP	66	55639 → 80 [ACK] Seq=153426 Ack=778 Win=138944 Len=0 TSval=725607691 TSecr=3913851542

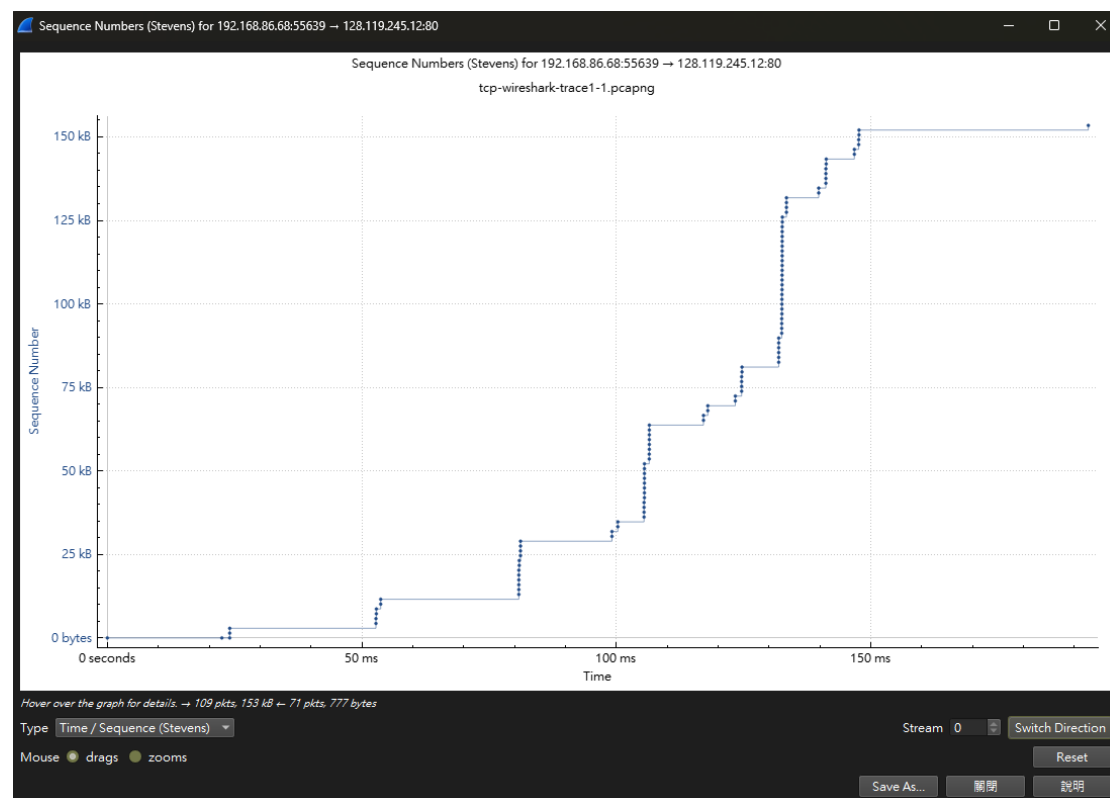
Part 4. TCP congestion control in action (第 11~12 題)

11. Use the *Time-Sequence-Graph(Stevens)* plotting tool to view the sequence number versus time plot of segments being sent from the client to the gaia.cs.umass.edu server. Consider the “fleets” of packets sent around $t = 0.025$, $t = 0.053$, $t = 0.082$ and $t = 0.1$. Comment on whether this looks as if TCP is in its slow start phase, congestion avoidance phase or some other phase.

TCP 在 Slow Start 階段

理由：

- 每個 RTT 後，發送的 packet 數量翻倍增長 ($3 \rightarrow 4 \rightarrow 8 \rightarrow 11$)
- Congestion Window (cwnd) 快速增加
- 這正是 slow start 的特徵：每收到一個 ACK，cwnd 增加 1 MSS
- 如果是 congestion avoidance，增長會更緩慢、線性



12. These “fleets” of segments appear to have some periodicity. What can you say about the period?

Fleet 1: 時間 0.024047

Fleet 2: 時間 0.052774 → 間隔 = 0.028727 秒

Fleet 3: 時間 0.080768 → 間隔 = 0.027994 秒

Fleet 4: 時間 0.105445 → 間隔 = 0.024677 秒

平均周期 $\approx 0.027-0.028$ 秒

周期性 $\approx 27-28$ 毫秒，接近 RTT (Round-Trip Time)

原因：每個 fleet 之間的時間間隔就是 RTT。當 client 發送完一批 packets 後，需要等待約 RTT 時間才能收到 ACK，然後觸發下一批 packets 發送。